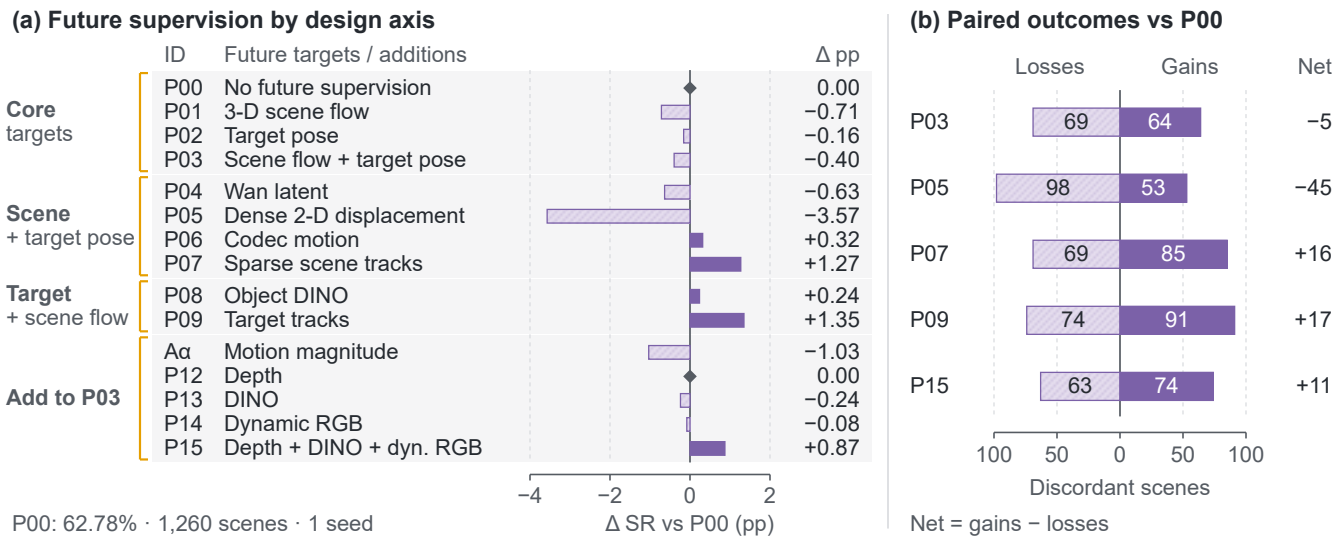

(a) Future supervision by design axis
ID
Future targets / additions
Δ pp
Core targets
P00 No future supervision 0.00
P01 3-D scene flow −0.71
P02 Target pose −0.16
P03 Scene flow + target pose −0.40
Scene + target pose
P04 Wan latent −0.63
P05 Dense 2-D displacement −3.57
P06 Codec motion +0.32
P07 Sparse scene tracks +1.27
Target + scene flow
P08 Object DINO +0.24
P09 Target tracks +1.35
Add to P03
Aα Motion magnitude −1.03
P12 Depth 0.00
P13 DINO −0.24
P14 Dynamic RGB −0.08
P15 Depth + DINO + dyn. RGB +0.87
−4
−2
0
2
P00: 62.78% · 1,260 scenes · 1 seed
Δ SR vs P00 (pp)
(b) Paired outcomes vs P00
Losses
Gains
Net
P03 69 64 −5
P05 98 53 −45
P07 69 85 +16
P09 74 91 +17
P15 63 74 +11
100
50
0
50
100
Discordant scenes
Net = gains − losses